TCV-ϕ V: cohesive quantum microstructure from CC-Φ_1 and a controlled bridge to CC-Φ_2 flavour templates

Cyrille LECROQ[1]

[1] *Independent Researcher*

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This fifth paper develops the quantum-microstructure layer of the canonical TCV-ϕ programme. Our scope is to formulate a reproducible CC-Φ_1 cluster framework, quantify its curvature response, and connect it to coarse-grained effective quantities used in previous papers. We also include a clearly labelled exploratory bridge to CC-Φ_2 flavour toy models, without claiming a complete theory-level fit of flavour observables. All claims are restricted to script-backed diagnostics archived in the Paper V workspace.

I. INTRODUCTION

Paper V follows Papers I–IV and keeps the same canonical EFT field content and notation. The focus is the microscopic cohesive layer (CC-Φ_1) and its controlled link to exploratory CC-Φ_2 flavour templates.

II. CANONICAL EFT BASELINE

We keep exactly the same action and parameter hierarchy as in Papers I–IV. No new fundamental fields or couplings are introduced in this paper.

III. LOCAL CC-Φ_1 CLUSTERS: SETUP

We adopt a minimal cluster quantization toy consistent with the CC-Φ_1 preprint construction of Ref. [5]:

$$L_c(R) \sim \frac{1}{m_{\rm eff}(R)}, \qquad m_{\rm eff}^2(R) = m_1^2 + \xi_0 R, \tag{1}$$

and

$$\Xi_n(R) = m_{\rm eff}^2(R) + \left(\frac{n\pi}{L_c(R)}\right)^2. \tag{2}$$

Here Ξ_n has mass-squared units, while the corresponding mode energy scale is

$$\omega_n(R) = \sqrt{\Xi_n(R)}. \tag{3}$$

The baseline benchmark used by `compute_ccphi1_spectrum.py` is $m_1 = 10^{-8}$, $\xi_0 = 0.1$, and $n_{\rm modes} = 3$.

IV. CC-Φ_1 SPECTRUM AND CURVATURE RESPONSE

From `ccphi1_spectrum.json`, the baseline values at $R = 0$ are

$$\Xi_1 \simeq 1.09 \times 10^{-15}, \qquad \Xi_2 - \Xi_1 \simeq 2.96 \times 10^{-15}, \tag{4}$$

which corresponds to $\omega_1(0) \simeq 3.30 \times 10^{-8}$ and coherence length $L_c(0) \simeq 10^8$ in Planck units. The curvature scan (`compute_ccphi1_curvature_response.py`) uses $0 \le R \le 2 \times 10^{-15}$ and yields

$$\frac{\Xi_1(R)}{\Xi_1(0)} \in [1, 3], \qquad \frac{L_c(R)}{L_c(0)} \in [0.577, 1]. \tag{5}$$

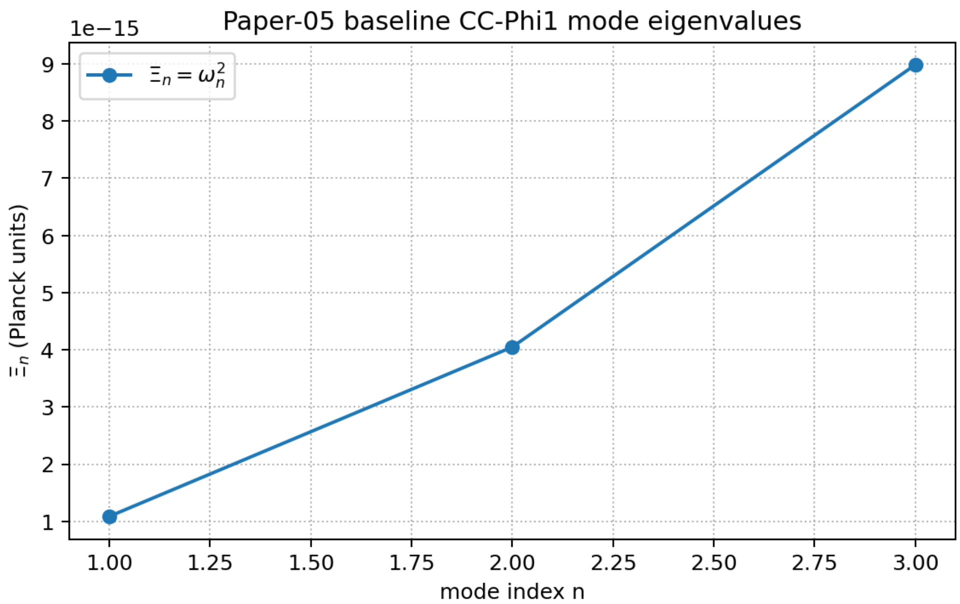

FIG. 1. Baseline CC-Φ_1 mode spectrum from `compute_ccphi1_spectrum.py`.

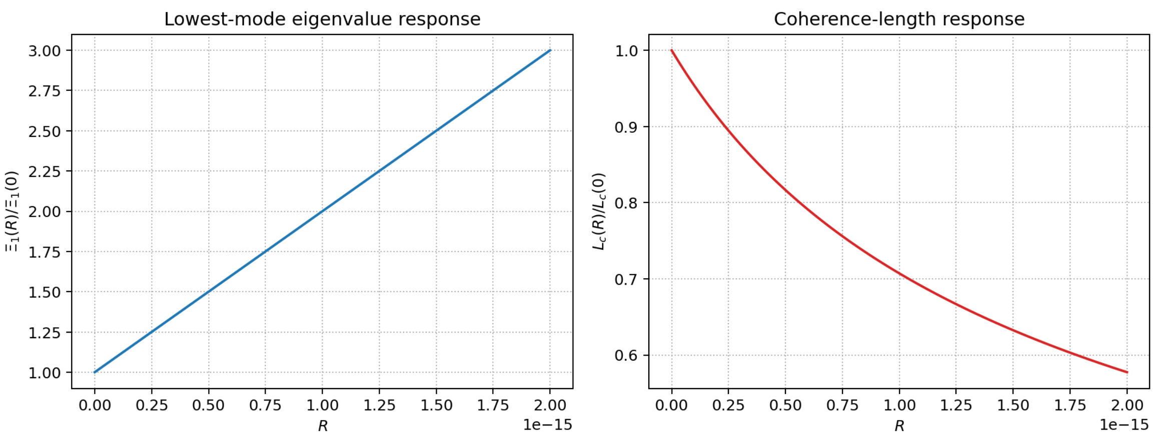

FIG. 2. Curvature response of the lowest mode and coherence length from `compute_ccphi1_curvature_response.py`.

V. COARSE-GRAINING AND EFFECTIVE MACROSCOPIC CONSISTENCY

At the current toy level we use a proxy density $n_{\rm CC} \sim m_1^3$ and define

$$\rho_{\rm cc,eff} \sim n_{\rm CC}\,\omega_1, \qquad p_{\rm proxy} = w_{\rm proxy}\rho_{\rm cc,eff}, \tag{6}$$

with the illustrative closure choice $w_{\rm proxy} = -1/2$ in the current script benchmark. This $w_{\rm proxy}$ is a toy coarse-graining parameter and should not be interpreted as the final EFT vacuum equation of state. The script `compute_ccphi1_coarse_grain.py` gives

$$\rho_{\rm cc,eff} \in [3.30, 5.71] \times 10^{-32}, \tag{7}$$

over the same curvature window. These values are used as consistency diagnostics, not as a final microscopic QFT derivation.

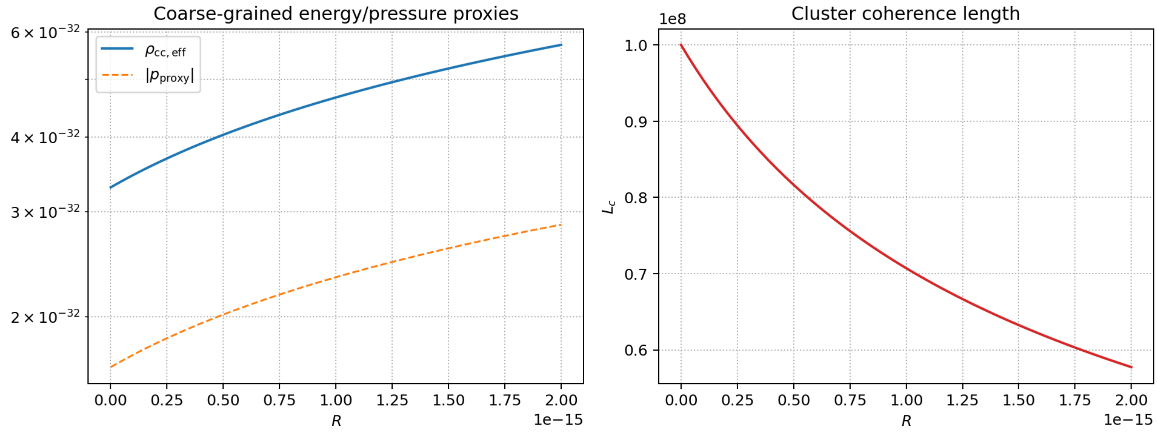

FIG. 3. Toy coarse-graining diagnostics from `compute_ccphi1_coarse_grain.py`.

Parameter	Benchmark value
m_2	10^{-2}
L_c	50
N_{modes}	5
ϵ_ℓ	0.007
ϵ_ν	0.20
κ_ν	1.5

TABLE I. Exploratory CC-Φ_2 flavour-toy benchmark used in `compute_ccphi2_flavour_toy.py`.

VI. EXPLORATORY BRIDGE TO CC-Φ_2 FLAVOUR SECTOR

We reuse the preprint-X flavour toy logic ([6]) through an in-repo module for full reproducibility in `compute_ccphi2_flavour_to`
The toy mass matrices are

$$M_\ell = \mathrm{diag}(m_e, m_\mu, m_\tau) + \epsilon_\ell\, C_\ell^\dagger H_{\mathrm{modes}} C_\ell^T, \tag{8}$$

$$M_\nu = \mathrm{diag}(m_{\nu_1}, m_{\nu_2}, m_{\nu_3}) + \epsilon_\nu\, C_\nu^\dagger H_{\mathrm{modes}} C_\nu^T, \tag{9}$$

with $U_{\mathrm{PMNS}} = U_\ell^\dagger U_\nu$ after diagonalization. Here H_{modes} denotes the CC-Φ_2 mode-space Hamiltonian and C_ℓ, C_ν are flavour-projection coefficient matrices in that mode basis; their explicit toy forms are given in Ref. [6] and mirrored in the in-repo module. For the representative run we obtain PMNS-like angles

$$\theta_{12} \simeq 21.1^\circ, \quad \theta_{13} \simeq 8.73^\circ, \quad \theta_{23} \simeq 30.4^\circ, \tag{10}$$

with $\delta_{\mathrm{CP}} \simeq 59^\circ$ in the same toy setup. Relative to current global-fit central values (roughly $\theta_{12} \sim 33^\circ$, $\theta_{13} \sim 8.6^\circ$, $\theta_{23} \sim 49^\circ$), this toy keeps only an order-of-magnitude flavour hierarchy and is not a precision fit. No precision global-fit claim is made at this stage.

VII. STATUS OF ASSUMPTIONS

- EFT-derived baseline definitions inherited from Papers I–IV.
- Phenomenological closures used in the CC-Φ_1 cluster pipeline.
- Exploratory templates for CC-Φ_2 flavour mapping.

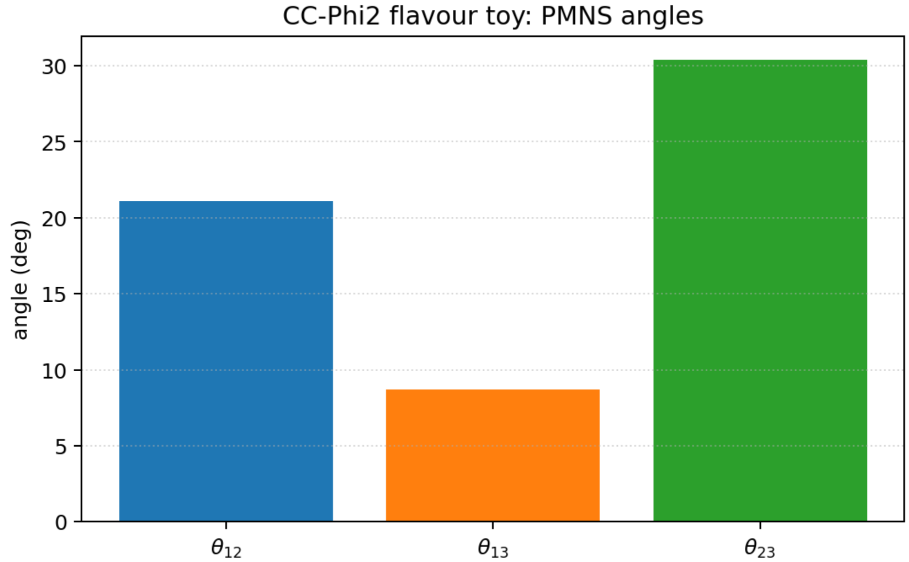

FIG. 4. Exploratory CC-Φ_2 flavour toy angles from reused preprint-X code.

Block	Output artifact	Status label
CC-Φ_1 spectrum	`ccphi1_spectrum.json`	phenomenological toy
Curvature response	`ccphi1_curvature_scan.npz`	phenomenological scan
Coarse-graining	`ccphi1_coarse_grain.json`	toy proxy
CC-Φ_2 flavour	`ccphi2_flavour_toy.json`	exploratory toy

TABLE II. Paper-V script-backed outputs and status labels.

VIII. CONCLUSION

Paper V is designed as a reproducible and conservative microstructure step in the TCV-ϕ series. It prepares a falsifiable bridge between cohesive quantum microstructure and later high-level sectors.

[1] C. Lecroq, "TCV-ϕ I," preprint (2026).
[2] C. Lecroq, "TCV-ϕ II," preprint (2026).
[3] C. Lecroq, "TCV-ϕ III," preprint (2026).
[4] C. Lecroq, "TCV-ϕ IV," preprint (2026).
[5] C. Lecroq, "Cohesive Quantum Gravity and CC-Φ_1 microstructure," preprint (2026).
[6] C. Lecroq, "From Quantum CC-Φ_2 to flavour structures," preprint (2026).